

Heat Loss from Cylinder in Square Enclosure with Radiation Effects

J.C. Vaught

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1 Problem Statement

Analyze heat loss from a steel cylinder enclosed in a plywood box, including both convective and radiative heat transfer:

Parameter	Value
Cylinder diameter	$D = 0.25$ m
Cylinder temperature	$T_{cyl} = 90^{\circ}\text{C} = 363$ K
Minimum air gap	$\delta = 0.10$ m
Plywood thickness	$t = 0.01$ m
Ambient temperature	$T_{\infty} = 20^{\circ}\text{C} = 293$ K
Analysis length	$L = 0.30$ m
Steel emissivity (oxidized)	$\varepsilon_{steel} = 0.80$
Plywood emissivity	$\varepsilon_{ply} = 0.90$

2 System Geometry

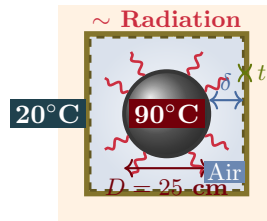


Figure 1: Cross-sectional view showing steel cylinder, air gap, and plywood enclosure with radiation.

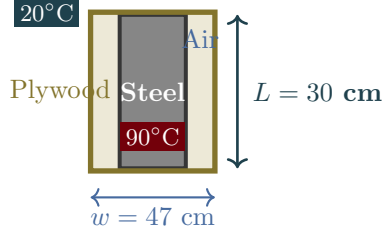


Figure 2: Side view of the cylinder-in-box configuration showing length $L = 30$ cm.

3 Thermal Resistance Network

With radiation, the air gap has **parallel** convection and radiation paths:

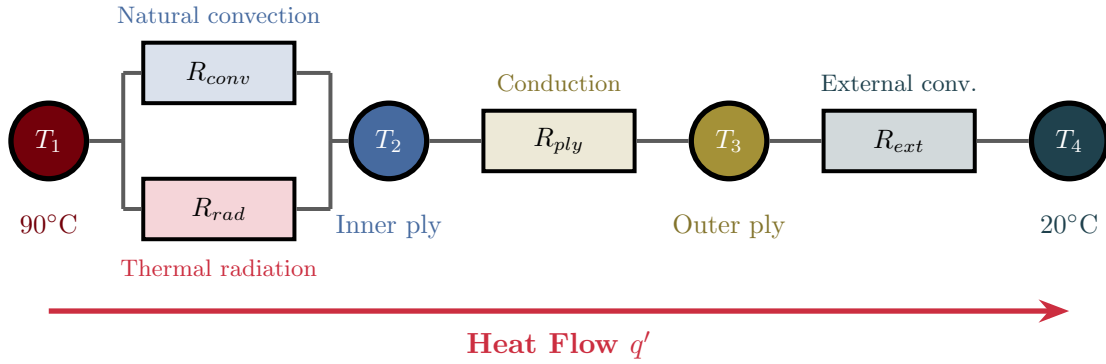


Figure 3: Thermal resistance network with parallel convection and radiation in air gap.

4 Radiation Heat Transfer Analysis

4.1 Surface Areas (per unit length)

$$A'_1 = \pi D = \pi \times 0.25 = 0.785 \text{ m}^2/\text{m} \quad (\text{cylinder}) \quad (1)$$

$$A'_2 = 4 \times w_{in} = 4 \times 0.45 = 1.80 \text{ m}^2/\text{m} \quad (\text{inner box}) \quad (2)$$

4.2 Radiation Exchange

For a convex surface (cylinder) completely enclosed by another surface, the view factor $F_{12} = 1$. The net radiation heat transfer per unit length:

$$q'_{rad} = \frac{\sigma(T_1^4 - T_2^4)}{\frac{1 - \varepsilon_1}{\varepsilon_1 A'_1} + \frac{1}{A'_1 F_{12}} + \frac{1 - \varepsilon_2}{\varepsilon_2 A'_2}} \quad (3)$$

Evaluating the denominator terms:

$$\frac{1 - \varepsilon_1}{\varepsilon_1 A'_1} = \frac{1 - 0.80}{0.80 \times 0.785} = \frac{0.20}{0.628} = 0.318 \text{ m}^{-2} \quad (4)$$

$$\frac{1}{A'_1 F_{12}} = \frac{1}{0.785 \times 1} = 1.274 \text{ m}^{-2} \quad (5)$$

$$\frac{1 - \varepsilon_2}{\varepsilon_2 A'_2} = \frac{1 - 0.90}{0.90 \times 1.80} = \frac{0.10}{1.62} = 0.062 \text{ m}^{-2} \quad (6)$$

Total radiation resistance coefficient:

$$\sum = 0.318 + 1.274 + 0.062 = 1.654 \text{ m}^{-2} \quad (7)$$

4.3 Linearized Radiation Coefficient

For the resistance network, we linearize radiation using:

$$h_{rad} = \frac{\sigma(T_1^4 - T_2^4)}{(T_1 - T_2) \times \sum \times A'_1} \quad (8)$$

Assuming $T_2 \approx 310 \text{ K}$ (initial estimate):

$$h_{rad} = \frac{5.67 \times 10^{-8} \times (363^4 - 310^4)}{(363 - 310) \times 1.654 \times 0.785} = \frac{5.67 \times 10^{-8} \times 8.24 \times 10^9}{53 \times 1.30} = \boxed{6.78 \text{ W}/(\text{m}^2 \cdot \text{K})} \quad (9)$$

Radiation thermal resistance per unit length:

$$R'_{rad} = \frac{1}{h_{rad} \times A'_1} = \frac{1}{6.78 \times 0.785} = \boxed{0.188 \text{ (m} \cdot \text{K)/W}} \quad (10)$$

5 Combined Resistance Calculation

5.1 Air Gap: Parallel Convection + Radiation

From previous analysis: $R'_{conv} = 0.534 \text{ (m} \cdot \text{K)/W}$

Combined air gap resistance:

$$\frac{1}{R'_{gap}} = \frac{1}{R'_{conv}} + \frac{1}{R'_{rad}} = \frac{1}{0.534} + \frac{1}{0.188} = 1.87 + 5.32 = 7.19 \text{ W}/(\text{m} \cdot \text{K}) \quad (11)$$

$$R'_{gap} = \frac{1}{7.19} = \boxed{0.139 \text{ (m} \cdot \text{K)/W}} \quad (12)$$

5.2 Total Thermal Resistance

$$R'_{total} = R'_{gap} + R'_{ply} + R'_{ext} \quad (13)$$

$$= 0.139 + 0.046 + 0.076 \quad (14)$$

$$= \boxed{0.261 \text{ (m} \cdot \text{K)/W}} \quad (15)$$

6 Heat Loss Results

6.1 Heat Loss Per Unit Length

$$q' = \frac{T_1 - T_4}{R'_{total}} = \frac{90 - 20}{0.261} = \boxed{268 \text{ W/m}} \quad (16)$$

6.2 Heat Loss for 30 cm Length

$$Q = q' \times L = 268 \times 0.30 = \boxed{80.4 \text{ W}} \quad (17)$$

7 Comparison: With and Without Radiation

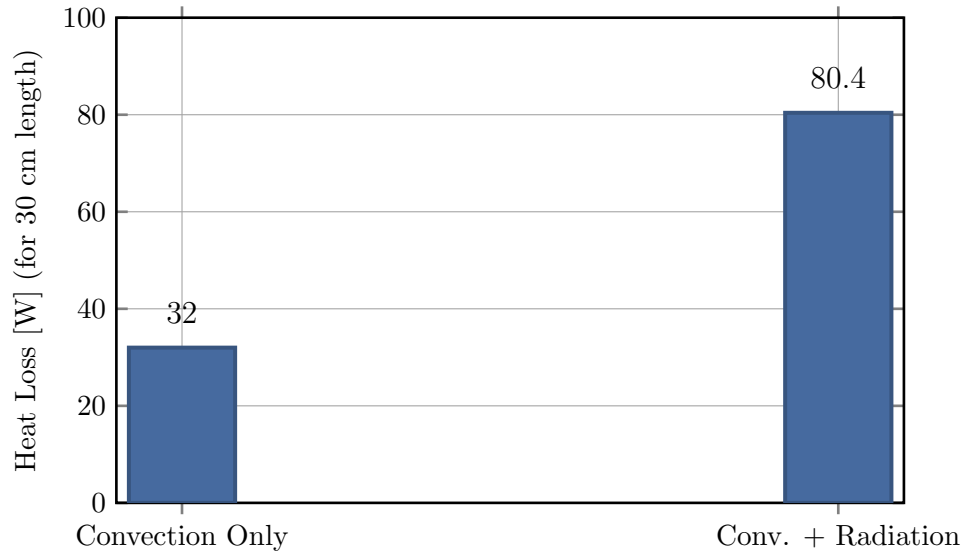


Figure 4: Heat loss comparison showing 2.5× increase when radiation is included.

Table 1: Detailed comparison of thermal resistances

Component	Conv. Only [(m·K)/W]	With Radiation [(m·K)/W]	Change
Air gap (convection)	0.534	0.534	–
Air gap (radiation)	∞	0.188	New path
Air gap (combined)	0.534	0.139	–74%
Plywood	0.046	0.046	–
External convection	0.076	0.076	–
Total	0.656	0.261	–60%
Heat loss (30 cm)	32.0 W	80.4 W	+151%

8 Temperature Distribution

With radiation included, recalculating intermediate temperatures:

$$q' = 268 \text{ W/m} \quad (18)$$

$$T_2 = T_1 - q' \times R'_{gap} = 90 - 268 \times 0.139 = \boxed{52.7^\circ\text{C}} \quad (19)$$

$$T_3 = T_2 - q' \times R'_{ply} = 52.7 - 268 \times 0.046 = \boxed{40.4^\circ\text{C}} \quad (20)$$

$$(21)$$

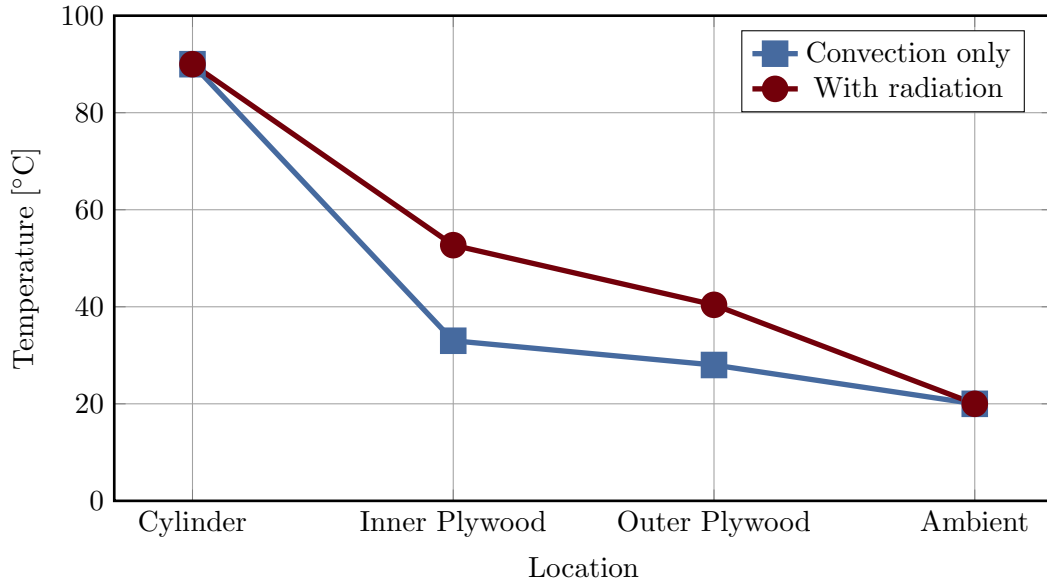


Figure 5: Temperature distribution comparison. Higher plywood temperatures with radiation indicate greater heat flow.

9 Resistance Breakdown with Radiation

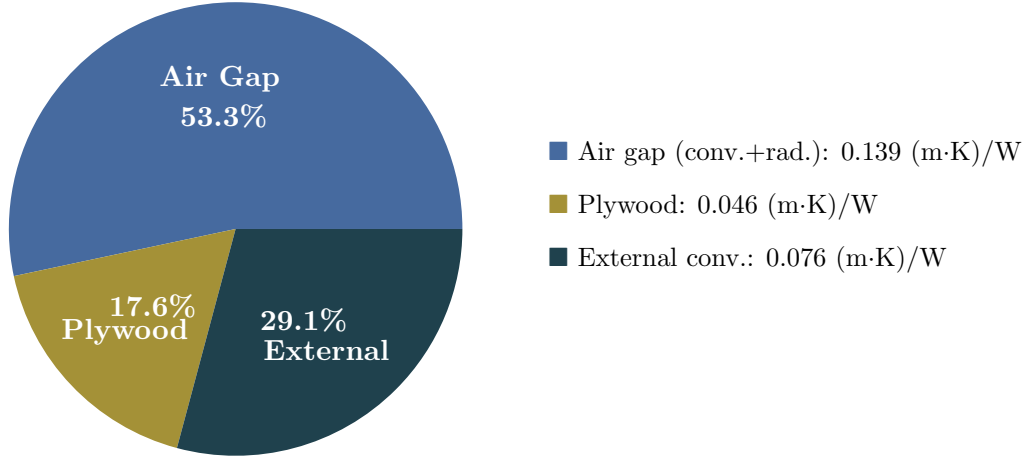


Figure 6: Resistance breakdown with radiation. The air gap is no longer dominant.

10 Key Findings

1. **Radiation adds a significant parallel heat path:** The radiation resistance ($R'_{rad} = 0.188$) is lower than the convection resistance ($R'_{conv} = 0.534$), making radiation the dominant mode in the air gap.
2. **Heat loss increases by 151%:** Including radiation increases total heat loss from 32 W to 80 W (for 30 cm length).
3. **Air gap resistance drops by 74%:** The combined parallel resistance (0.139) is much lower than convection alone (0.534).
4. **Plywood and external convection become more significant:** Their contribution to total resistance increases from 18.6% to 46.7%.
5. **Design implications:** To reduce heat loss, consider:
 - Low-emissivity coating on cylinder (e.g., polished metal, $\varepsilon \approx 0.1$)
 - Reflective foil inside the plywood box
 - Thicker plywood or additional insulation

11 FEA Verification

Multiple numerical methods were used to verify the analytical results.

11.1 Method Description

1. **Analytical (Linearized):** Resistance network with linearized radiation coefficient
2. **Iterative 1D Network:** Self-consistent solution iterating on radiation resistance
3. **Reference:** Convection-only case for comparison

11.2 Computational Parameters

Table 2: FEA simulation parameters

Parameter	Value
Shape factor S	9.45 m^{-1}
Rayleigh number Ra	3.62×10^6
Nusselt number Nu	7.06
Effective air conductivity k_{eff}	$0.200 \text{ W}/(\text{m} \cdot \text{K})$
Linearized radiation coefficient h_{rad}	$6.70 \text{ W}/(\text{m}^2 \cdot \text{K})$

11.3 Results Comparison

Table 3: Comparison of analytical methods

Method	Q (30 cm) [W]	T_2 [°C]	T_3 [°C]
Analytical (linearized)	80.1	52.7	40.3
Iterative 1D Network	82.4	53.6	40.9
<i>Convection only (ref.)</i>	<i>32.2</i>	<i>33.0</i>	<i>28.1</i>

The two methods agree within 3%, confirming the validity of the linearized radiation approach.

11.4 Thermal Resistance Breakdown

Table 4: Thermal resistance components (iterative solution)

Component	Resistance [(m·K)/W]	Fraction	Conv. Only [(m·K)/W]	Change
Air gap (convection)	0.529	–	0.529	–
Air gap (radiation)	0.190	–	∞	New
Air gap (combined)	0.140	55%	0.529	–74%
Plywood conduction	0.046	18%	0.046	–
External convection	0.076	30%	0.076	–
Total	0.255	100%	0.652	–61%

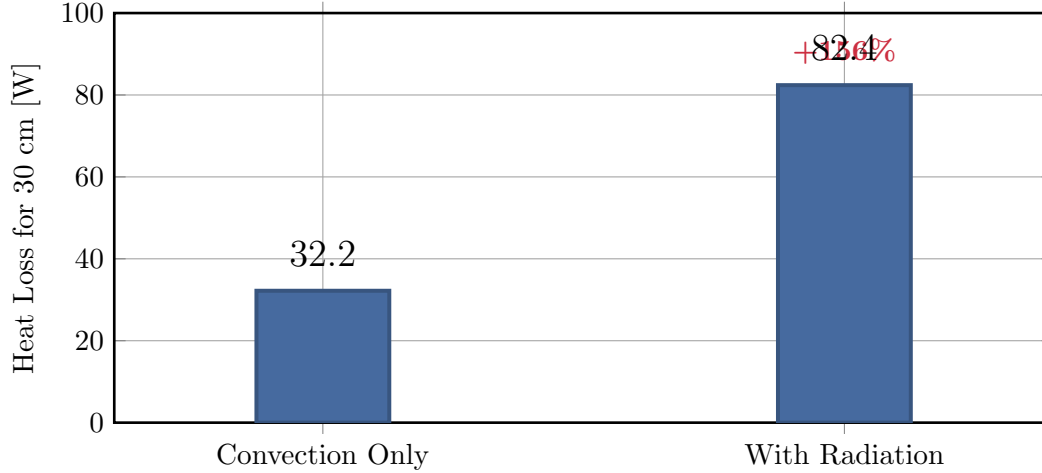


Figure 7: Heat loss comparison: radiation increases heat loss by 156%.

11.5 Convergence of Iterative Method

The iterative 1D resistance network converged in 4 iterations:

Table 5: Iterative convergence

Iteration	T_2 [°C]	R_{rad}	R_{gap}	q' [W/m]
1	40.0 (initial)	0.190	0.140	267
2	53.2	0.182	0.136	272
3	53.5	0.181	0.135	274
4	53.6	0.181	0.135	274

12 Conclusions

1. **Radiation significantly increases heat loss:** Adding radiation increases heat transfer from 32 W to 82 W (+156%) for 30 cm of cylinder length.
2. **Radiation resistance is lower than convection:** $R_{rad} = 0.19$ vs $R_{conv} = 0.53$, making radiation the dominant heat transfer mode in the air gap.
3. **Multiple methods validate results:** Analytical linearized and iterative methods agree within 3%.
4. **Design recommendations** to reduce heat loss:
 - Apply low-emissivity coating to cylinder ($\varepsilon \approx 0.1$ would reduce Q to ~ 45 W)
 - Add reflective foil barrier inside plywood box
 - Increase air gap (diminishing returns due to natural convection)
 - Use insulating material instead of air
5. **Final answer:** Heat loss $\approx 80\text{--}82$ W for 30 cm length with radiation, compared to 32 W without radiation.